

MATAM, Senegal

WMO: 616300

Lat: 15.630N

Lon: 13.250W

Elev: 50

StdP: 14.67

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	60.6	62.8	29.6	23.7	78.6	32.4	26.9	76.8	15.4	79.4	13.2	79.7	1.6	40	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	24.7	111.7	76.3	109.9	75.7	108.0	75.3	83.7	96.3	82.8	95.1	82.1	94.2	3.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	160.2	90.0	80.0	156.0	89.2	79.2	151.7	88.5	47.7	96.1	46.7	95.2	45.9	94.3	97.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.1	9.9	8.4	DB	54.7	115.6	2.4	1.2	53.0	116.4	51.6	117.1	50.2	117.8	48.4	118.7
			WB	45.9	87.0	1.7	3.5	44.6	89.5	43.6	91.5	42.7	93.4	41.4	96.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	87.7	77.8	82.1	88.3	93.5	96.6	94.9	89.5	86.6	87.0	90.4	86.2	79.9
	DBStd	6.68	3.89	4.49	4.47	3.49	3.26	3.92	3.73	3.37	3.24	2.98	3.15	3.99
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	13777	861	899	1187	1305	1444	1348	1225	1134	1110	1253	1085	926
	CDD65	8302	396	479	722	855	979	898	760	669	660	788	635	461
	CDH74	119871	4219	5785	10364	13828	16667	14937	11407	9068	8967	11461	8206	4962
	CDH80	76421	2172	3329	6685	9670	12228	10658	7073	4880	4904	7159	4962	2701
Wind	WSAvg	3.3	2.9	3.4	3.1	3.4	4.1	4.9	4.6	3.6	3.1	2.4	2.0	2.6
Precipitation	PrecAvg	14.7	0.0	0.0	0.0	0.0	0.0	1.1	3.6	5.4	3.8	0.7	0.0	0.1
	PrecMax	27.7	0.2	0.9	0.1	0.2	0.9	4.3	8.6	12.4	11.7	4.6	1.8	1.7
	PrecMin	6.5	0.0	0.0	0.0	0.0	0.0	0.0	0.4	1.2	1.1	0.0	0.0	0.0
	PrecStd	5.0	0.0	0.2	0.0	0.0	0.2	1.1	2.0	2.4	2.2	1.0	0.2	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.3	105.0	109.7	112.9	114.6	113.2	105.9	102.2	102.2	107.3	105.5	101.9
		MCWB	68.9	71.8	73.0	75.9	75.7	76.8	77.6	79.7	81.5	77.3	73.1	71.5
	2%	DB	96.2	101.5	107.6	110.7	112.5	110.3	102.7	98.7	99.1	104.8	103.4	98.7
		MCWB	67.1	69.1	72.9	75.6	76.6	77.1	78.9	80.4	81.7	77.7	72.8	69.3
	5%	DB	93.5	98.9	105.5	108.9	110.8	107.8	100.5	96.6	97.1	103.0	101.3	96.1
		MCWB	65.8	67.9	71.9	74.6	76.2	77.6	78.9	80.8	81.6	77.7	72.2	68.5
	10%	DB	91.2	96.1	102.6	106.7	108.5	105.4	98.4	94.6	95.1	100.8	99.1	93.1
		MCWB	64.8	66.6	70.8	73.7	75.6	77.3	78.6	80.6	81.5	77.6	71.7	67.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.8	75.4	76.9	82.0	82.8	83.7	83.2	83.9	85.0	84.5	79.2	74.1
		MCDB	93.6	98.5	103.3	104.4	105.3	101.6	96.7	95.0	96.4	97.7	96.3	98.0
	2%	WB	69.5	72.2	75.2	78.5	80.6	82.0	82.1	82.8	83.7	82.9	77.1	71.7
		MCDB	92.6	96.5	103.4	106.6	105.4	100.8	95.5	93.7	94.7	97.0	95.5	95.5
	5%	WB	67.3	69.9	73.6	76.7	79.1	80.8	81.3	82.2	82.9	81.8	75.2	69.7
		MCDB	91.0	94.2	101.9	105.4	104.7	99.6	94.4	92.7	93.3	95.7	94.3	93.4
	10%	WB	65.4	67.9	71.8	75.2	77.7	79.7	80.5	81.5	82.2	80.8	73.5	67.8
		MCDB	88.9	93.2	99.6	103.5	103.0	98.6	93.4	91.4	92.1	94.1	94.4	90.8
Mean Daily Temperature Range	5% DB	MDBR	27.2	27.4	28.1	27.0	24.7	20.8	16.9	15.2	16.1	21.7	27.6	27.3
		MCDBR	31.2	30.6	31.1	29.1	27.4	23.6	19.2	17.7	18.5	24.9	30.2	31.6
		MCWBR	14.3	13.5	13.5	12.5	10.8	8.2	5.9	5.9	6.0	8.2	11.5	13.4
	5% WB	MCDBR	29.3	28.1	28.6	26.5	23.5	20.7	17.2	16.2	16.8	20.5	26.2	29.6
		MCWBR	14.4	13.7	13.7	12.6	10.3	7.9	6.4	6.3	7.0	7.9	11.2	13.6
Clear-Sky Solar Irradiance	taub		0.535	0.562	0.638	0.660	0.712	0.777	0.741	0.652	0.619	0.606	0.515	0.507
	taud		1.789	1.733	1.592	1.597	1.509	1.467	1.559	1.733	1.788	1.784	1.907	1.890
	Ebn at Noon		239	240	226	222	207	192	200	220	228	226	243	242
	Edh at Noon		67	74	87	87	94	96	88	75	71	69	59	59
All-Sky Solar Radiation	RadAvg		1650	1856	2113	2230	2204	2081	1970	1881	1892	1838	1674	1571
	RadStd		96	89	83	75	79	78	62	74	68	64	73	61

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page